

**Trigonometric Identities**

Trigonometric identities are equations that express relations among trigonometric functions which are true for all values of the variables involved.

1. Reciprocal Relations

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

2. Even-odd Identities

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\cot(-\theta) = -\cot \theta$$

$$\sec(-\theta) = \sec \theta$$

$$\csc(-\theta) = -\csc \theta$$

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3. Co-function Identities

$$\sin \theta = \cos(90 - \theta)$$

$$\cos \theta = \sin(90 - \theta)$$

$$\tan \theta = \cot(90 - \theta)$$

$$\cot \theta = \tan(90 - \theta)$$

$$\sec \theta = \csc(90 - \theta)$$

$$\csc \theta = \sec(90 - \theta)$$

4. Pythagorean Relations

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

5. Sum of Angles Formulas

$$\sin(\theta + \alpha) = \sin \theta \cos \alpha + \cos \theta \sin \alpha$$

$$\cos(\theta + \alpha) = \cos \theta \cos \alpha - \sin \theta \sin \alpha$$

$$\tan(\theta + \alpha) = \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha}$$

6. Difference of Angles Formula

$$\sin(\theta - \alpha) = \sin \theta \cos \alpha - \cos \theta \sin \alpha$$

$$\cos(\theta - \alpha) = \cos \theta \cos \alpha + \sin \theta \sin \alpha$$

$$\tan(\theta - \alpha) = \frac{\tan \theta - \tan \alpha}{1 + \tan \theta \tan \alpha}$$

7. Double Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

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8. Power of Functions

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$$

$$\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

9. Functions of Half-Angles

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta}$$

10. Sum of Two Functions

$$\sin \theta + \sin \alpha = 2 \sin \frac{1}{2}(\theta + \alpha) \cos \frac{1}{2}(\theta - \alpha)$$

$$\cos \theta + \cos \alpha = 2 \cos \frac{1}{2}(\theta + \alpha) \cos \frac{1}{2}(\theta - \alpha)$$

$$\tan \theta + \tan \alpha = \frac{\sin(\theta + \alpha)}{\cos \theta \cos \alpha}$$

11. Difference of Two Functions

$$\sin \theta - \sin \alpha = 2 \cos \frac{1}{2}(\theta + \alpha) \sin \frac{1}{2}(\theta - \alpha)$$

$$\cos \theta - \cos \alpha = 2 \sin \frac{1}{2}(\theta + \alpha) \sin \frac{1}{2}(\theta - \alpha)$$

$$\tan \theta - \tan \alpha = \frac{\sin(\theta - \alpha)}{\cos \theta \cos \alpha}$$

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12. Product of Two Functions

$$2 \sin \theta \sin \alpha = \cos(\theta - \alpha) - \cos(\theta + \alpha)$$

$$2 \sin \theta \cos \alpha = \sin(\theta + \alpha) + \sin(\theta - \alpha)$$

$$2 \cos \theta \cos \alpha = \cos(\theta + \alpha) + \cos(\theta - \alpha)$$

Exponential Forms of Trigonometric Functions

The following are the trigonometric functions expressed in terms of exponential functions and the imaginary unit.

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

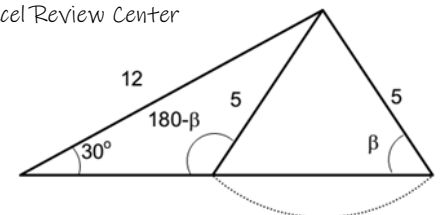
$$\tan x = \frac{-i(e^{ix} - e^{-ix})}{e^{ix} + e^{-ix}}$$

$$\cot x = \frac{i(e^{ix} + e^{-ix})}{e^{ix} - e^{-ix}}$$

Ambiguous case

If you are given a triangle with two sides and an opposite angle known, you can solve one of the two unknown angles using the law of sines. Since $\sin \theta = \sin(180 - \theta)$, a unique solution is obtained only in right triangles. When dealing with oblique triangles, you will find two triangles. This phenomenon is known as the ambiguous case

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$$\sin \beta = \sin(180 - \beta)$$

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